

SMART LENDER – SAMPLE PROJECT DOCUMENTATION

Project Overview

Project Title: Smart Lender – Loan Approval Prediction System

Domain: Machine Learning | Banking | Finance

Technology Stack:

- Python
- Flask
- Scikit-learn
- Pandas
- NumPy
- HTML
- CSS
- JavaScript

Project Description

Smart Lender is a machine learning-based web application developed to predict whether a loan application should be approved or rejected. The system analyzes applicant information such as income, education, employment status, loan amount, credit history, marital status, and property area to generate accurate predictions.

The application reduces manual verification efforts and helps financial institutions make faster and more reliable lending decisions.

Project Objectives

- Predict loan approval using Machine Learning.
- Reduce manual verification time.
- Improve loan approval accuracy.
- Provide an easy-to-use web application.
- Support quick decision-making for banks.

The project uses a Loan Prediction dataset containing applicant information.

Important Features

- Gender
- Married
- Dependents
- Education

- Self Employed
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Amount Term
- Credit History
- Property Area
- Loan Status (Target Variable)

Folder Structure

```
SMART-LENDER/
├── Dataset/
│   └── loan_prediction.csv
├── Flask/
│   ├── static/
│   │   ├── css/
│   │   ├── images/
│   │   └── js/
│   ├── templates/
│   │   ├── home.html
│   │   ├── input.html
│   │   └── output.html
│   ├── app.py
│   ├── rdf.pkl
│   └── scaler.pkl
├── IBM/
├── Training/
│   └── Loan Prediction using ML.ipynb
├── .gitignore
├── LICENSE
├── README.md
└── requirements.txt
```

Dataset Preview

The dataset is stored inside the Dataset folder as loan_prediction.csv.

It contains historical loan applicant records that are used to train and evaluate the machine learning model.

Loan Approval Prediction

Gender

Male



Married

Yes



Dependents

Education

Graduate



Self Employed

Yes



Applicant Income

Coapplicant Income

Loan Amount

Loan Amount Term

Model Training

The model training process was performed using Jupyter Notebook.

The notebook includes:

- Data Loading
- Data Cleaning
- Feature Encoding
- Data Scaling
- Model Training
- Model Evaluation
- Model Saving

The trained model is saved as:

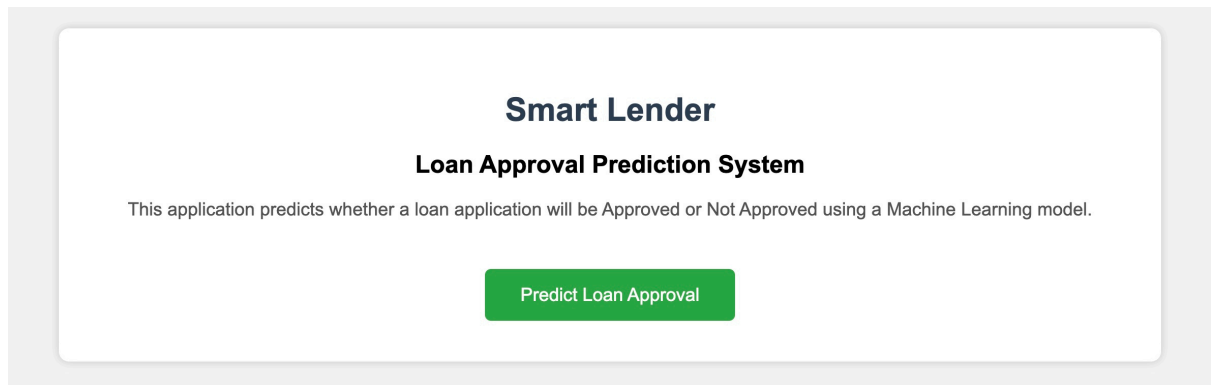
- rdf.pkl
- scaler.pkl

Web Application

The Flask web application provides a simple interface for predicting loan approval.

Main Pages:

- Home Page
- Loan Input Form
- Prediction Result Page



Loan Prediction Form

Users enter applicant details including:

- Gender
- Married Status
- Dependents
- Education
- Self Employment
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Term
- Credit History
- Property Area

The information is submitted to the trained model for prediction.

Prediction Result

After processing the user inputs, the machine learning model predicts whether the loan will be approved.

The prediction result is displayed immediately on the result page.

Loan Approval Prediction

Loan Will be Approved ✓

Predict Another Loan

Project Execution

The Flask application is executed locally using : python app.py

The application runs at : <http://127.0.0.1:5000>

```
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 984-851-120
```

Conclusion

Smart Lender successfully demonstrates how machine learning can automate loan approval prediction. The Flask-based web application provides real-time predictions using a trained Random Forest model, making the process faster, more consistent, and more efficient. The project highlights the practical use of machine learning in banking and financial decision-making.